

**UNITED STATES DISTRICT COURT  
DISTRICT OF NEVADA**

|                                                                                                   |                                                                                                                                                                                                             |
|---------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MARK JOSEPH NEJMEH,<br><br>Plaintiff,<br><br>v.<br><br>THERIVA BIOLOGICS, INC.,<br><br>Defendant. | <b>Case No. 3:26-cv-00705</b><br><br><b>PROPOSED SUPPLEMENTAL INFORMATION</b><br><br><b>PART N—EXHIBITS</b><br><b>PUBLICLY IDENTIFIED U.S. PATENT PORTFOLIO</b><br><b>AND REQUEST FOR COMPLETE REGISTER</b> |
|---------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**Proposed Patent Exhibit Schedule and Disclosure-Gap Request**

*Organizational appendix for review before filing; not a patent-title opinion*

**Purpose and limitation.** This appendix organizes the U.S. issued-grant records located in the reviewed public sources under Theriva Biologics, Inc., its predecessor name Synthetic Biologics, Inc., and its subsidiary VCN Biosciences, S.L. It identifies 53 U.S. issued grants. It does not purport to be the complete worldwide portfolio, a current chain-of-title report, or proof that every listed right remains owned, enforceable, maintained, unencumbered, or commercially valuable. It also does not equate patent language with FDA approval or clinical efficacy.

**Why a complete register is still required.** Theriva's 2025 Form 10-K reports more than 135 U.S. and foreign patents and more than 50 pending applications and states that VCN controls more than 50 patents through assignment or exclusive license, but it names only examples. Public assignee indexes are useful locating tools, not certified title evidence. A complete answer therefore requires the Company's program-by-program register, assignment records, maintenance records, and every governing license. The gap between aggregate counts and an itemized current register is a question for production and testimony, not proof of concealment.

**Ownership categories.** An assigned patent and an exclusively licensed patent are different interests. The SYN-004 record is described by Theriva as assigned to Theriva. The SYN-020 record is also described as assigned to Theriva, but the 10-K states that all such patents were exclusively licensed to Rasayana in February 2026. VCN's portfolio includes patents assigned to VCN and patents exclusively licensed from third parties. Those categories are kept separate below.

**Exhibit P-1 — What Theriva's 2025 Form 10-K discloses**

| Portfolio statement                                                                                                               | Named examples                                                                                                                                                                   | What remains unverified publicly                                                                                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Theriva reports more than 135 U.S. and foreign patents and more than 50 pending applications.                                     | The filing names nine U.S. SYN-004 patents as examples: 8,894,994; 9,587,234; 9,301,995; 9,301,996; 9,290,754; 9,376,673; 9,404,103; 9,464,280; and 9,695,409.                   | Every patent/application number, jurisdiction, family, current owner, status, expiration, maintenance history, encumbrance, and license term.                                                          |
| SYN-020 is supported by U.S. and foreign patents and applications assigned to Theriva.                                            | The filing describes formulations, medical uses, and manufacture, with expected expiration in 2038–2040, and states all such patents were licensed to Rasayana in February 2026. | The complete Rasayana patent schedule, present chain of title, retained rights, sublicenses, maintenance duties, security interests, valuation, and any later amendment.                               |
| VCN controls more than 50 U.S. and foreign patents through assignment or exclusive license and more than 15 pending applications. | The filing names U.S. 10,316,065 as an example and identifies licenses from IDIBELL, ICO, and Hospital Sant Joan de Deu.                                                         | Which rights are assigned to VCN versus third-party-owned and licensed; the complete family schedule; current license compliance; royalties; reversion; maintenance; and change-of-control provisions. |

## Exhibit P-2 — U.S. grants located under Theriva Biologics, Inc. (14)

These are public issued-grant records located under the current company name. Their appearance in an assignee index does not replace an official current title and maintenance review. The 2025 Form 10-K states that the SYN-020 patents were licensed to Rasayana in February 2026.

| Ex.   | Patent and title                                                                    | Dates                                          | Program, record name, and inventors                                                                                                                                                                                                                                                                                         |
|-------|-------------------------------------------------------------------------------------|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P-2.1 | U.S. 12,648,987<br>Alkaline phosphatase agents for treatment of radiation disorders | Filed: Apr. 14, 2023<br>Issued: June 9, 2026   | Program: SYN-020 / IAP<br>Record: Theriva Biologics assignee record<br>Inventor(s): Michael Kaleko; Christian Furlan Freguia; Sheila Connelly<br>Link: <a href="https://patents.google.com/patent/US12648987">https://patents.google.com/patent/US12648987</a>                                                              |
| P-2.2 | U.S. 12,508,304<br>Alkaline phosphatase formulations                                | Filed: June 17, 2024<br>Issued: Dec. 30, 2025  | Program: SYN-020 / IAP<br>Record: Theriva Biologics assignee record<br>Inventor(s): Vincent Wachter; J. Blair West; Michael Kaleko; Christian Furlan Freguia<br>Link: <a href="https://patents.google.com/patent/US12508304">https://patents.google.com/patent/US12508304</a>                                               |
| P-2.3 | U.S. 12,472,147<br>Intestinal alkaline phosphatase formulations                     | Filed: Mar. 17, 2023<br>Issued: Nov. 18, 2025  | Program: SYN-020 / IAP<br>Record: Theriva Biologics assignee record<br>Inventor(s): Andrew Bristol; Ray Stapleton; Michael Kaleko; Christian Furlan Freguia; Steven Hubert; Cristina Freire; James Gubbins<br>Link: <a href="https://patents.google.com/patent/US12472147">https://patents.google.com/patent/US12472147</a> |
| P-2.4 | U.S. 12,318,434<br>Alkaline phosphate-based oncology treatments                     | Filed: May 5, 2020<br>Issued: June 3, 2025     | Program: SYN-020 / IAP<br>Record: Theriva Biologics assignee record<br>Inventor(s): Christian Furlan Freguia; Michael Kaleko<br>Link: <a href="https://patents.google.com/patent/US12318434">https://patents.google.com/patent/US12318434</a>                                                                               |
| P-2.5 | U.S. 12,274,740<br>Microbiome protection from oral antibiotics                      | Filed: Dec. 11, 2023<br>Issued: Apr. 15, 2025  | Program: SYN-004 / microbiome<br>Record: Theriva Biologics assignee record<br>Inventor(s): Michael Kaleko; Sheila Connelly<br>Link: <a href="https://patents.google.com/patent/US12274740">https://patents.google.com/patent/US12274740</a>                                                                                 |
| P-2.6 | U.S. 12,178,917<br>Beta-lactamase formulations and uses thereof                     | Filed: Sept. 28, 2020<br>Issued: Dec. 31, 2024 | Program: SYN-004 / ribaxamase<br>Record: Theriva Biologics assignee record<br>Inventor(s): Andrew Bristol; Michael Kaleko; Sheila Connelly<br>Link: <a href="https://patents.google.com/patent/US12178917">https://patents.google.com/patent/US12178917</a>                                                                 |
| P-2.7 | U.S. 12,048,735<br>Alkaline phosphatase formulations                                | Filed: Mar. 29, 2021<br>Issued: July 30, 2024  | Program: SYN-020 / IAP<br>Record: Theriva Biologics assignee record<br>Inventor(s): Vincent Wachter; J. Blair West; Michael Kaleko; Christian Furlan Freguia<br>Link: <a href="https://patents.google.com/patent/US12048735">https://patents.google.com/patent/US12048735</a>                                               |
| P-2.8 | U.S. 11,981,899<br>E. coli-based production of beta-lactamase                       | Filed: Dec. 1, 2022<br>Issued: May 14, 2024    | Program: SYN-004 / manufacture<br>Record: Theriva Biologics assignee record<br>Inventor(s): Andrew Bristol; Michael Kaleko; Steven Hubert<br>Link: <a href="https://patents.google.com/patent/US11981899">https://patents.google.com/patent/US11981899</a>                                                                  |

| Ex.    | Patent and title                                                                                            | Dates                                         | Program, record name, and inventors                                                                                                                                                                                                                                                                                            |
|--------|-------------------------------------------------------------------------------------------------------------|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P-2.9  | U.S. 11,944,669<br>Method and compositions for inhibiting or preventing adverse effects of oral antibiotics | Filed: Feb. 2, 2023<br>Issued: Apr. 2, 2024   | Program: SYN-004 / microbiome<br>Record: Theriva Biologics assignee record<br>Inventor(s): Michael Kaleko; Sheila Connelly; Vincent John Wachter<br>Link:<br><a href="https://patents.google.com/patent/US11944669">https://patents.google.com/patent/US11944669</a>                                                           |
| P-2.10 | U.S. 11,883,474<br>Microbiome protection from oral antibiotics                                              | Filed: Oct. 26, 2022<br>Issued: Jan. 30, 2024 | Program: SYN-004 / microbiome<br>Record: Theriva Biologics assignee record<br>Inventor(s): Michael Kaleko; Sheila Connelly<br>Link:<br><a href="https://patents.google.com/patent/US11883474">https://patents.google.com/patent/US11883474</a>                                                                                 |
| P-2.11 | U.S. 11,872,268<br>Safe and effective beta-lactamase dosing for microbiome protection                       | Filed: Jan. 10, 2022<br>Issued: Jan. 16, 2024 | Program: SYN-004 / dosing<br>Record: Theriva Biologics assignee record<br>Inventor(s): Joseph Sliman<br>Link:<br><a href="https://patents.google.com/patent/US11872268">https://patents.google.com/patent/US11872268</a>                                                                                                       |
| P-2.12 | U.S. 11,654,184<br>Alkaline phosphatase agents for treatment of radiation disorders                         | Filed: Mar. 20, 2019<br>Issued: May 23, 2023  | Program: SYN-020 / IAP<br>Record: Theriva Biologics assignee record<br>Inventor(s): Michael Kaleko; Christian Furlan Freguia; Sheila Connelly<br>Link:<br><a href="https://patents.google.com/patent/US11654184">https://patents.google.com/patent/US11654184</a>                                                              |
| P-2.13 | U.S. 11,638,699<br>Intestinal alkaline phosphatase formulations                                             | Filed: Mar. 20, 2019<br>Issued: May 2, 2023   | Program: SYN-020 / IAP<br>Record: Theriva Biologics assignee record<br>Inventor(s): Andrew Bristol; Ray Stapleton; Michael Kaleko; Christian Furlan Freguia; Steven Hubert; Cristina Freire; James Gubbins<br>Link:<br><a href="https://patents.google.com/patent/US11638699">https://patents.google.com/patent/US11638699</a> |
| P-2.14 | U.S. 11,608,494<br>Beta-lactamases with improved properties for therapy                                     | Filed: Dec. 8, 2021<br>Issued: Mar. 21, 2023  | Program: SYN-004 / ribaxamase<br>Record: Theriva Biologics assignee record<br>Inventor(s): Michael Kaleko; Sheila Connelly<br>Link:<br><a href="https://patents.google.com/patent/US11608494">https://patents.google.com/patent/US11608494</a>                                                                                 |

### Exhibit P-3 — U.S. grants located under Synthetic Biologics or otherwise tied to the predecessor record (38)

Synthetic Biologics changed its name to Theriva Biologics in October 2022. The entries below preserve the name shown in the public locating source. Some are co-assigned with another institution. No conclusion about current ownership, exclusivity, lapse, abandonment, or encumbrance should be drawn without the official chain of title and maintenance record.

| Ex.   | Patent and title                                               | Dates                                        | Program, public record, and inventors                                                                                                                                                                                                                  |
|-------|----------------------------------------------------------------|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P-3.1 | U.S. 11,542,510<br>E. coli-based production of beta-lactamase  | Filed: May 7, 2021<br>Issued: Jan. 3, 2023   | Program: SYN-004 / manufacture<br>Record: Synthetic Biologics record<br>Inventor(s): Andrew Bristol; Michael Kaleko; Steven Hubert<br>Link:<br><a href="https://patents.google.com/patent/US11542510">https://patents.google.com/patent/US11542510</a> |
| P-3.2 | U.S. 11,517,614<br>Microbiome protection from oral antibiotics | Filed: June 28, 2017<br>Issued: Dec. 6, 2022 | Program: SYN-004 / microbiome<br>Record: Synthetic Biologics record<br>Inventor(s): Michael Kaleko; Sheila Connelly<br>Link:<br><a href="https://patents.google.com/patent/US11517614">https://patents.google.com/patent/US11517614</a>                |

| Ex.    | Patent and title                                                                                            | Dates                                         | Program, public record, and inventors                                                                                                                                                                                                                                     |
|--------|-------------------------------------------------------------------------------------------------------------|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P-3.3  | U.S. 11,338,020<br>Alkaline phosphatase agents for treatment of neurodevelopmental disorders                | Filed: Jan. 8, 2019<br>Issued: May 24, 2022   | Program: SYN-020 / IAP<br>Record: Synthetic Biologics record<br>Inventor(s): Michael Kaleko; Christian Furlan Freguia; Sheila Connelly<br>Link:<br><a href="https://patents.google.com/patent/US11338020">https://patents.google.com/patent/US11338020</a>                |
| P-3.4  | U.S. 11,253,577<br>Safe and effective beta-lactamase dosing for microbiome protection                       | Filed: June 4, 2020<br>Issued: Feb. 22, 2022  | Program: SYN-004 / dosing<br>Record: Synthetic Biologics record<br>Inventor(s): Joseph Sliman<br>Link:<br><a href="https://patents.google.com/patent/US11253577">https://patents.google.com/patent/US11253577</a>                                                         |
| P-3.5  | U.S. 11,236,319<br>Beta-lactamases with improved properties for therapy                                     | Filed: July 24, 2020<br>Issued: Feb. 1, 2022  | Program: SYN-004 / ribaxamase<br>Record: Google patent record; 2023 name-change entry<br>Inventor(s): Michael Kaleko; Sheila Connelly<br>Link:<br><a href="https://patents.google.com/patent/US11236319">https://patents.google.com/patent/US11236319</a>                 |
| P-3.6  | U.S. 11,034,966<br>E. coli-based production of beta-lactamase                                               | Filed: Aug. 27, 2015<br>Issued: June 15, 2021 | Program: SYN-004 / manufacture<br>Record: Synthetic Biologics record<br>Inventor(s): Andrew Bristol; Michael Kaleko; Steven Hubert<br>Link:<br><a href="https://patents.google.com/patent/US11034966">https://patents.google.com/patent/US11034966</a>                    |
| P-3.7  | U.S. 10,987,410<br>Alkaline phosphatase formulations                                                        | Filed: July 17, 2018<br>Issued: Apr. 27, 2021 | Program: SYN-020 / IAP<br>Record: Synthetic Biologics record<br>Inventor(s): Vincent Wachter; J. Blair West; Michael Kaleko; Christian Furlan Freguia<br>Link:<br><a href="https://patents.google.com/patent/US10987410">https://patents.google.com/patent/US10987410</a> |
| P-3.8  | U.S. 10,828,260<br>Beta-lactamase formulations and uses thereof                                             | Filed: Aug. 24, 2018<br>Issued: Nov. 10, 2020 | Program: SYN-004 / ribaxamase<br>Record: Synthetic Biologics record<br>Inventor(s): Andrew Bristol; Michael Kaleko; Sheila Connelly<br>Link:<br><a href="https://patents.google.com/patent/US10828260">https://patents.google.com/patent/US10828260</a>                   |
| P-3.9  | U.S. 10,792,346<br>Method and compositions for inhibiting or preventing adverse effects of oral antibiotics | Filed: July 10, 2018<br>Issued: Oct. 6, 2020  | Program: SYN-004 / microbiome<br>Record: Synthetic Biologics record<br>Inventor(s): Michael Kaleko; Sheila Connelly; Vincent John Wachter<br>Link:<br><a href="https://patents.google.com/patent/US10792346">https://patents.google.com/patent/US10792346</a>             |
| P-3.10 | U.S. 10,767,171<br>Beta-lactamases with improved properties for therapy                                     | Filed: Jan. 24, 2020<br>Issued: Sept. 8, 2020 | Program: SYN-004 / ribaxamase<br>Record: Synthetic Biologics record<br>Inventor(s): Michael Kaleko; Sheila Connelly<br>Link:<br><a href="https://patents.google.com/patent/US10767171">https://patents.google.com/patent/US10767171</a>                                   |
| P-3.11 | U.S. 10,736,871<br>Anti-methanogenic lovastatin analogs or derivatives and uses thereof                     | Filed: Mar. 31, 2016<br>Issued: Aug. 11, 2020 | Program: Legacy / IBS<br>Record: Cedars-Sinai and Synthetic Biologics co-assignee record<br>Inventor(s): Mark Pimentel; Vince Wachter<br>Link:<br><a href="https://patents.google.com/patent/US10736871">https://patents.google.com/patent/US10736871</a>                 |
| P-3.12 | U.S. 10,709,773<br>Safe and effective beta-lactamase dosing for microbiome protection                       | Filed: Mar. 7, 2016<br>Issued: July 14, 2020  | Program: SYN-004 / dosing<br>Record: Synthetic Biologics record<br>Inventor(s): Joseph Sliman<br>Link:<br><a href="https://patents.google.com/patent/US10709773">https://patents.google.com/patent/US10709773</a>                                                         |
| P-3.13 | U.S. 10,584,326<br>Beta-lactamases with improved properties for therapy                                     | Filed: May 16, 2019<br>Issued: Mar. 10, 2020  | Program: SYN-004 / ribaxamase<br>Record: Synthetic Biologics record<br>Inventor(s): Michael Kaleko; Sheila Connelly<br>Link:<br><a href="https://patents.google.com/patent/US10584326">https://patents.google.com/patent/US10584326</a>                                   |

| Ex.    | Patent and title                                                                                             | Dates                                         | Program, public record, and inventors                                                                                                                                                                                                                                                                           |
|--------|--------------------------------------------------------------------------------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P-3.14 | U.S. 10,548,955<br>Carbapenemases for use with antibiotics for the protection of the intestinal microbiome   | Filed: Jan. 9, 2018<br>Issued: Feb. 4, 2020   | Program: Microbiome / beta-lactamase<br>Record: Synthetic Biologics record<br>Inventor(s): Michael Kaleko; Sheila Connelly<br>Link:<br><a href="https://patents.google.com/patent/US10548955">https://patents.google.com/patent/US10548955</a>                                                                  |
| P-3.15 | U.S. 10,526,400<br>Humanized pertussis antibodies and uses thereof                                           | Filed: June 25, 2018<br>Issued: Jan. 7, 2020  | Program: Legacy / pertussis<br>Record: University of Texas System and Synthetic Biologics co-assignee record<br>Inventor(s): Jennifer Maynard; Annalee Nguyen; Eduardo Padlan; Ellen Wagner<br>Link:<br><a href="https://patents.google.com/patent/US10526400">https://patents.google.com/patent/US10526400</a> |
| P-3.16 | U.S. 10,336,995<br>Beta-lactamases with improved properties for therapy                                      | Filed: Aug. 24, 2018<br>Issued: July 2, 2019  | Program: SYN-004 / ribaxamase<br>Record: Synthetic Biologics record<br>Inventor(s): Michael Kaleko; Sheila Connelly<br>Link:<br><a href="https://patents.google.com/patent/US10336995">https://patents.google.com/patent/US10336995</a>                                                                         |
| P-3.17 | U.S. 10,253,306<br>Modified beta-lactamases and methods and uses related thereto                             | Filed: July 27, 2017<br>Issued: Apr. 9, 2019  | Program: SYN-004 / ribaxamase<br>Record: Synthetic Biologics record<br>Inventor(s): Pertti Koski; Ulla Airaksinen; Katja Valimäki<br>Link:<br><a href="https://patents.google.com/patent/US10253306">https://patents.google.com/patent/US10253306</a>                                                           |
| P-3.18 | U.S. 10,105,322<br>Beta-lactamase formulations and uses thereof                                              | Filed: Oct. 8, 2015<br>Issued: Oct. 23, 2018  | Program: SYN-004 / ribaxamase<br>Record: Synthetic Biologics record<br>Inventor(s): Andrew Bristol; Michael Kaleko; Sheila Connelly<br>Link:<br><a href="https://patents.google.com/patent/US10105322">https://patents.google.com/patent/US10105322</a>                                                         |
| P-3.19 | U.S. 10,087,433<br>Beta-lactamases with improved properties for therapy                                      | Filed: May 30, 2018<br>Issued: Oct. 2, 2018   | Program: SYN-004 / ribaxamase<br>Record: Synthetic Biologics record<br>Inventor(s): Michael Kaleko; Sheila Connelly<br>Link:<br><a href="https://patents.google.com/patent/US10087433">https://patents.google.com/patent/US10087433</a>                                                                         |
| P-3.20 | U.S. 10,046,035<br>Methods and compositions for inhibiting or preventing adverse effects of oral antibiotics | Filed: July 5, 2017<br>Issued: Aug. 14, 2018  | Program: SYN-004 / microbiome<br>Record: Synthetic Biologics record<br>Inventor(s): Michael Kaleko; Sheila Connelly; Vincent John Wachter<br>Link:<br><a href="https://patents.google.com/patent/US10046035">https://patents.google.com/patent/US10046035</a>                                                   |
| P-3.21 | U.S. 10,041,056<br>Modified beta-lactamases and methods and uses related thereto                             | Filed: Jan. 27, 2017<br>Issued: Aug. 7, 2018  | Program: SYN-004 / ribaxamase<br>Record: Synthetic Biologics record<br>Inventor(s): Pertti Koski; Ulla Airaksinen; Katja Valimäki<br>Link:<br><a href="https://patents.google.com/patent/US10041056">https://patents.google.com/patent/US10041056</a>                                                           |
| P-3.22 | U.S. 10,035,846<br>Humanized pertussis antibodies and uses thereof                                           | Filed: Nov. 18, 2016<br>Issued: July 31, 2018 | Program: Legacy / pertussis<br>Record: University of Texas System and Synthetic Biologics co-assignee record<br>Inventor(s): Jennifer Maynard; Annalee Nguyen; Eduardo Padlan; Ellen Wagner<br>Link:<br><a href="https://patents.google.com/patent/US10035846">https://patents.google.com/patent/US10035846</a> |
| P-3.23 | U.S. 10,011,824<br>Beta-lactamases with improved properties for therapy                                      | Filed: Aug. 10, 2017<br>Issued: July 3, 2018  | Program: SYN-004 / ribaxamase<br>Record: Synthetic Biologics record<br>Inventor(s): Michael Kaleko; Sheila Connelly<br>Link:<br><a href="https://patents.google.com/patent/US10011824">https://patents.google.com/patent/US10011824</a>                                                                         |

| Ex.    | Patent and title                                                                                           | Dates                                          | Program, public record, and inventors                                                                                                                                                                                                                                                                         |
|--------|------------------------------------------------------------------------------------------------------------|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P-3.24 | U.S. 9,956,292<br>Anti-methanogenic compositions and uses thereof                                          | Filed: Aug. 13, 2015<br>Issued: May 1, 2018    | Program: Legacy / IBS<br>Record: Cedars-Sinai and Synthetic Biologics co-assignee record<br>Inventor(s): Mark Pimentel; Ruchi Mathur; Steve Kanzer; Vince Wachter<br>Link:<br><a href="https://patents.google.com/patent/US9956292">https://patents.google.com/patent/US9956292</a>                           |
| P-3.25 | U.S. 9,783,797<br>Beta-lactamases with improved properties for therapy                                     | Filed: June 2, 2017<br>Issued: Oct. 10, 2017   | Program: SYN-004 / ribaxamase<br>Record: Synthetic Biologics record<br>Inventor(s): Michael Kaleko; Sheila Connelly<br>Link:<br><a href="https://patents.google.com/patent/US9783797">https://patents.google.com/patent/US9783797</a>                                                                         |
| P-3.26 | U.S. 9,765,320<br>Modified beta-lactamases and methods and uses related thereto                            | Filed: Apr. 26, 2016<br>Issued: Sept. 19, 2017 | Program: SYN-004 / ribaxamase<br>Record: Synthetic Biologics record<br>Inventor(s): Pertti Koski; Ulla Airaksinen; Katja Valimaki<br>Link:<br><a href="https://patents.google.com/patent/US9765320">https://patents.google.com/patent/US9765320</a>                                                           |
| P-3.27 | U.S. 9,744,221<br>Method and compositions for inhibiting or preventing adverse effects of oral antibiotics | Filed: Dec. 23, 2015<br>Issued: Aug. 29, 2017  | Program: SYN-004 / microbiome<br>Record: Synthetic Biologics record<br>Inventor(s): Michael Kaleko; Sheila Connelly; Vincent John Wachter<br>Link:<br><a href="https://patents.google.com/patent/US9744221">https://patents.google.com/patent/US9744221</a>                                                   |
| P-3.28 | U.S. 9,695,409<br>Beta-lactamases with improved properties for therapy                                     | Filed: Aug. 24, 2016<br>Issued: July 4, 2017   | Program: SYN-004 / ribaxamase<br>Record: Synthetic Biologics record<br>Inventor(s): Michael Kaleko; Sheila Connelly<br>Link:<br><a href="https://patents.google.com/patent/US9695409">https://patents.google.com/patent/US9695409</a>                                                                         |
| P-3.29 | U.S. 9,587,234<br>Modified beta-lactamases and methods and uses related thereto                            | Filed: Feb. 26, 2016<br>Issued: Mar. 7, 2017   | Program: SYN-004 / ribaxamase<br>Record: Synthetic Biologics record<br>Inventor(s): Pertti Koski; Ulla Airaksinen; Katja Valimaki<br>Link:<br><a href="https://patents.google.com/patent/US9587234">https://patents.google.com/patent/US9587234</a>                                                           |
| P-3.30 | U.S. 9,512,204<br>Humanized pertussis antibodies and uses thereof                                          | Filed: Mar. 31, 2015<br>Issued: Dec. 6, 2016   | Program: Legacy / pertussis<br>Record: University of Texas System and Synthetic Biologics co-assignee record<br>Inventor(s): Jennifer Maynard; Annalee Nguyen; Eduardo Padlan; Ellen Wagner<br>Link:<br><a href="https://patents.google.com/patent/US9512204">https://patents.google.com/patent/US9512204</a> |
| P-3.31 | U.S. 9,464,280<br>Beta-lactamases with improved properties for therapy                                     | Filed: July 1, 2016<br>Issued: Oct. 11, 2016   | Program: SYN-004 / ribaxamase<br>Record: Synthetic Biologics record<br>Inventor(s): Michael Kaleko; Sheila Connelly<br>Link:<br><a href="https://patents.google.com/patent/US9464280">https://patents.google.com/patent/US9464280</a>                                                                         |
| P-3.32 | U.S. 9,404,103<br>Beta-lactamases with improved properties for therapy                                     | Filed: May 20, 2016<br>Issued: Aug. 2, 2016    | Program: SYN-004 / ribaxamase<br>Record: Synthetic Biologics record<br>Inventor(s): Michael Kaleko; Sheila Connelly<br>Link:<br><a href="https://patents.google.com/patent/US9404103">https://patents.google.com/patent/US9404103</a>                                                                         |
| P-3.33 | U.S. 9,376,673<br>Beta-lactamases with improved properties for therapy                                     | Filed: Feb. 9, 2016<br>Issued: June 28, 2016   | Program: SYN-004 / ribaxamase<br>Record: Synthetic Biologics record<br>Inventor(s): Michael Kaleko; Sheila Connelly<br>Link:<br><a href="https://patents.google.com/patent/US9376673">https://patents.google.com/patent/US9376673</a>                                                                         |
| P-3.34 | U.S. 9,301,996<br>Modified beta-lactamases and methods and uses related thereto                            | Filed: Apr. 1, 2015<br>Issued: Apr. 5, 2016    | Program: SYN-004 / ribaxamase<br>Record: Synthetic Biologics record<br>Inventor(s): Pertti Koski; Ulla Airaksinen; Katja Valimaki<br>Link:<br><a href="https://patents.google.com/patent/US9301996">https://patents.google.com/patent/US9301996</a>                                                           |

| Ex.    | Patent and title                                                                | Dates                                         | Program, public record, and inventors                                                                                                                                                                                                                                 |
|--------|---------------------------------------------------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P-3.35 | U.S. 9,301,995<br>Modified beta-lactamases and methods and uses related thereto | Filed: Oct. 17, 2014<br>Issued: Apr. 5, 2016  | Program: SYN-004 / ribaxamase<br>Record: Synthetic Biologics record<br>Inventor(s): Pertti Koski; Ulla Airaksinen; Katja Valimaki<br>Link:<br><a href="https://patents.google.com/patent/US9301995">https://patents.google.com/patent/US9301995</a>                   |
| P-3.36 | U.S. 9,290,754<br>Beta-lactamases with improved properties for therapy          | Filed: Apr. 17, 2015<br>Issued: Mar. 22, 2016 | Program: SYN-004 / ribaxamase<br>Record: Form 10-K and Google patent record; 2023 name-change entry<br>Inventor(s): Michael Kaleko; Sheila Connelly<br>Link:<br><a href="https://patents.google.com/patent/US9290754">https://patents.google.com/patent/US9290754</a> |
| P-3.37 | U.S. 9,034,602<br>Modified beta-lactamases and methods and uses related thereto | Filed: May 17, 2011<br>Issued: May 19, 2015   | Program: SYN-004 / ribaxamase<br>Record: Synthetic Biologics record<br>Inventor(s): Pertti Koski; Ulla Airaksinen; Katja Valimaki<br>Link:<br><a href="https://patents.google.com/patent/US9034602">https://patents.google.com/patent/US9034602</a>                   |
| P-3.38 | U.S. 8,894,994<br>Modified beta-lactamases and methods and uses related thereto | Filed: Oct. 7, 2013<br>Issued: Nov. 25, 2014  | Program: SYN-004 / ribaxamase<br>Record: Synthetic Biologics record<br>Inventor(s): Pertti Koski; Ulla Airaksinen; Katja Valimaki<br>Link:<br><a href="https://patents.google.com/patent/US8894994">https://patents.google.com/patent/US8894994</a>                   |

#### Exhibit P-4 — U.S. grant located under VCN Biosciences, S.L. (1)

| Ex.   | Patent and title                                                                                                                 | Dates                                      | Program, public record, and inventors                                                                                                                                                                                                                          |
|-------|----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P-4.1 | U.S. 11,000,560<br>Oncolytic adenoviruses with mutations in immunodominant adenovirus epitopes and their use in cancer treatment | Filed: May 4, 2016<br>Issued: May 11, 2021 | Program: VCN oncolytic-virus platform<br>Record: VCN Biosciences, S.L. assignee record<br>Inventor(s): Ramon Alemany; Raul Gil; Miriam Bazan<br>Link:<br><a href="https://patents.google.com/patent/US1100560">https://patents.google.com/patent/US1100560</a> |

#### Exhibit P-5 — Disclosed controlled or licensed rights not represented as Theriva-owned

| Right                                                                     | Publicly disclosed ownership/control                                                                                                                                                               | Necessary limitation                                                                                                                                   |
|---------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| U.S. Patent No. 10,316,065, Oncolytic adenoviruses for treating cancer    | The public patent record identifies IDIBELL and Institut Catala d'Oncologia as assignees. Theriva's 10-K identifies the patent as an example within VCN's assigned-or-exclusively-licensed estate. | Treat as third-party-owned and licensed/controlled unless a later recorded assignment proves otherwise.                                                |
| Spanish patent P200901201, Oncolytic adenoviruses for treating cancer     | The 10-K states VCN obtained an exclusive license from IDIBELL in 2010 and that the patent is co-owned by IDIBELL and ICO.                                                                         | The license includes royalty, termination, utilization, and maintenance provisions; the full agreement and current compliance must be reviewed.        |
| Spanish patent P200700665 and EP-family priority application 14 38 2162.7 | The 10-K describes additional VCN licenses from ICO and IDIBELL/ICO.                                                                                                                               | Patent-family members, present owners, territories, status, payment duties, and change-of-control terms require the complete schedules and agreements. |

#### Exhibit P-6 — C. difficile patent and company-announcement chronology

| Date          | Documented event                                                                                                                                                                 | What it establishes—and does not                                                                                          |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Mar. 1, 2016  | Synthetic Biologics announced three USPTO Notices of Allowance covering SYN-004-related composition and methods, including prevention of CDI and antibiotic-associated diarrhea. | The Company publicized patent allowances. A notice of allowance is not an issued patent, FDA approval, or proof of value. |
| July 12, 2016 | The Company announced U.S. Patent No. 9,376,673 and a Notice of Allowance for U.S. application No. 15/160,669.                                                                   | The release identified the allowed application before issuance. It could not yet name the later final patent number.      |

| Date                    | Documented event                                                                                                                                                                                                           | What it establishes—and does not                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Aug. 2, 2016            | Application No. 15/160,669 issued as U.S. Patent No. 9,404,103 B1. Its summary states that the improved beta-lactamases have uses “including the prevention or treatment of CDI and/or a C. difficile-associated disease.” | This is patent language describing claimed technology. Prevention is a clinically meaningful therapeutic objective, but CDI is an infection that can cause colitis; it is not itself an ulcer. The patent does not call SYN-004 or CDI a “cure.” Its only “Cure” reference is the title of a cited article about a duodenal ulcer and Helicobacter pylori. Patent language is not FDA approval or proof of clinical efficacy. |
| Post-issuance search    | The reviewed Theriva/Synthetic release archive and web sources did not locate a separate company release after August 2, 2016 that names final U.S. Patent No. 9,404,103.                                                  | Absence from the reviewed sources is not proof that no communication existed. The Company should identify any release, investor communication, or decision record concerning post-issuance publicity.                                                                                                                                                                                                                         |
| Jan. 5 and May 11, 2017 | The Company later reported a 412-patient Phase 2b result and FDA Breakthrough Therapy Designation for prevention of CDI.                                                                                                   | Those later clinical/regulatory disclosures are separate from patent issuance and do not transform the patent itself into approval.                                                                                                                                                                                                                                                                                           |

### Exhibit P-7 — Complete C. difficile patent attached separately

The separate Exhibit P-7 PDF reproduces all 52 pages of U.S. Patent No. 9,404,103 B1. Yellow highlighting is limited to the two-line statement on patent page 2 that the improved beta-lactamases have therapeutic uses “including the prevention or treatment of CDI and/or a C. difficile-associated disease.” The exhibit cover explains that the patent does not use “cure” to describe SYN-004 or CDI and that patent language is not a regulatory approval or efficacy finding. No other complete patent is attached; the remaining patent records are supplied by number and public link in this schedule.

### Exhibit P-8 — Focused production needed to close the public-record gaps

- a complete patent-and-application register from January 1, 2016 to the present, organized by program and patent family, listing every jurisdiction, application and patent number, owner, beneficial owner, chain of title, inventors, priority, filing, issue, expiration, maintenance or annuity status, terminal disclaimer, lapse, abandonment, revival, security interest, assignment, license, sublicense, field restriction, territory, reversion right, and enforcement or challenge;
- the complete patent schedules, assignment instruments, recorded title documents, maintenance records, and amendments for the Rasayana license, the VCN acquisition, and every IDIBELL, ICO, Hospital Sant Joan de Deu, Cedars-Sinai, and University of Texas agreement reflected in the public record;
- all patent-specific valuations, impairment analyses, sale or license proposals, bidder or counterparty diligence, board or committee materials, and banker presentations used in any financing, restructuring, strategic review, merger, reverse merger, asset transfer, or license analysis;
- a program-by-program reconciliation showing which patents support SYN-004, SYN-020, VCN-01, Albumin Shield, or a discontinued legacy program, and identifying who paid prosecution and maintenance costs and where those costs were recorded; and
- the investor-relations and disclosure-decision record for patent allowances, issuances, lapses, abandonments, assignments, licenses, and sublicenses, including application No. 15/160,669 and U.S. Patent No. 9,404,103.

### Requested treatment and evidentiary guardrails

1. Use this schedule only as an organizational index to identified public records. A patent document may be capable of judicial notice for what it says and when it issued, but disputed ownership, maintenance, commercial value, materiality, and scientific effect require proper evidence and, where disputed, testimony.
2. If the Court finds the subject proportional and relevant to the pleaded warrant- and disclosure-related issues, Plaintiff requests production of the focused records above through the ordinary Rule 26 process or a specifically supported expedited order. This appendix is not itself a subpoena, title opinion, or proof of wrongdoing.
3. No adverse inference should be drawn merely because a patent number was not individually named in a Form 10-K, a press release was not located, a patent belongs to a continuation family, or a company used an exclusive license instead of ownership. The requested records are intended to determine the facts.



## Source record

1. [Theriva Form 10-K for the year ended Dec. 31, 2025, Intellectual Property and collaboration disclosures](#)
2. [Public assignee index — Theriva Biologics, Inc.](#)
3. [Public assignee index — Synthetic Biologics, Inc., page 1](#)
4. [Public assignee index — Synthetic Biologics, Inc., page 2](#)
5. [Public assignee index — Synthetic Biologics, Inc., page 3](#)
6. [Public assignee index — VCN Biosciences, S.L.](#)
7. [U.S. Patent No. 9,290,754 public patent record](#)
8. [U.S. Patent No. 11,236,319 public patent record](#)
9. [U.S. Patent No. 10,316,065 public patent record](#)
10. [USPTO MPEP § 301, ownership and assignability](#)
11. [U.S. Patent No. 9,404,103 B1 public patent text and downloadable PDF](#)
12. [Synthetic Biologics Mar. 1, 2016 SYN-004 patent-allowance release](#)
13. [Synthetic Biologics July 12, 2016 ribaxamase patent release](#)
14. [Synthetic Biologics Jan. 5, 2017 SEC-filed SYN-004 Phase 2b results release](#)
15. [Synthetic Biologics May 11, 2017 FDA Breakthrough Therapy Designation release](#)

**Respectfully submitted,**

Dated: \_\_\_\_\_

Signature: \_\_\_\_\_

**Mark Joseph Nejme, Plaintiff, self-represented**